ANDRY PAEZ

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RESEARCH EXPERIENCE

Research Assistant, Dept. of Physics and Astronomy **Feb 2026 – Present**

San Jose State University (advisor: Dr. Hilary Hurst), 30 hrs/week

- Develop spinor Bose-Einstein condensate simulations (1D Gross-Pitaevskii equation): imaginary-time ground-state preparation, time-splitting spectral (Strang) evolution, Bogoliubov-de Gennes spectra; all work merged through advisor-reviewed pull requests.
- Vectorized the sine-DVR kinetic-energy operator, achieving up to 1,880x speedup (161 s to 0.09 s at a 2048-point grid) with bit-exact numerical parity verified by a benchmark harness.
- Built the initial-condition pipeline: CLI with HDF5 output, hash-addressed run catalog with a rebuildable SQLite index, and a Slurm write-then-index workflow for the SJSU HPC cluster that avoids concurrent-writer contention.
- Designed controlled storage benchmarks (compression costs 55x write time for 11% disk savings) and authored the store-vs-recompute recommendation adopted as the project default.
- Validated spinor ground states to machine precision (norm error 2e-16) with correct miscible/immiscible domain-wall structure, and demonstrated the expected inverse-square-root scaling of domain-wall width with interaction strength.
- Implemented interaction-quench studies of spontaneous symmetry breaking with Truncated Wigner vacuum-noise seeding and domain-wall counting, extracting Kibble-Zurek defect scaling.
- Authored dated technical reports (benchmarks, validation, design recommendations) whose conclusions set project defaults.

PROJECTS

Quantum Portfolio Optimization, Womanium and WISER Program **Aug 2025**

- Implemented QAOA as a new ansatz family in an end-to-end portfolio optimization workflow (qiskit-algorithms, custom local search, Design-of-Experiments hyperparameter exposure); benchmarked against CPLEX on a 31-bond portfolio: median objective gap 5.6% with QAOA alone, 0.02% after local search.

Kalshi Weather-Market Trading System **Jan 2026 – Mar 2026**

- Built an end-to-end quantitative system: NWS and commercial forecast ingestion, Gaussian edge model, stacked LSTM temperature models (PyTorch) trained per city, and a Streamlit dashboard; documented in a LaTeX technical report with supporting analysis notebooks.

2D Ising Model Phase-Transition Study **Apr 2026 – May 2026**

- Built a Metropolis Monte Carlo simulation recovering the critical temperature (Tc of about 2.269 J/kB), with local energy-change optimization and small-lattice validation; documented in a LaTeX report and presentation.

OPEN SOURCE

- **hhurst/Quantum_Gases:** packaging (uv lockfile), CLI entry points, and cross-platform fixes merged into the research group's simulation codebase.
- **NousResearch/hermes-agent:** merged fixes for a systemd gateway restart race and Electron main-window lifecycle recovery in a large open-source agent framework.

SKILLS

- **Scientific Computing:** Python/NumPy, spectral and FFT methods, Monte Carlo, imaginary-time evolution, PyTorch, Qiskit, HDF5, Slurm/HPC, Jupyter, Matplotlib, LaTeX
- **Software Engineering:** FastAPI, pytest, GitHub Actions CI, Docker, SQLite, Git and PR-based review workflow, reproducible environments (uv), Linux
- **Also:** TypeScript/React, Rust (Tauri), C/C++, local LLM inference and agent tooling

EDUCATION

B.S., Physics **May 2026**

San Jose State University. GPA 3.80. Dean's Scholar (Fall 2024, Fall 2025)

A.S.-T Computer Science, Mathematics, Physics; A.A. Language and Rationality **May 2024**

Modesto Junior College. GPA 3.73. With Honors; President's List (5 terms)

Relevant Coursework: Computational Methods in Physics, Quantum Mechanics, Quantum Information Science, Mathematical Methods, Linear Algebra, Ordinary Differential Equations